SHORT-HORIZON TEMPORAL OPTICAL FLOW WITH PHYSICS-INFORMED CONSISTENCY

STATS-402: INTERDISCIPLINARY DATA ANALYSIS - FINAL REPORT

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ABSTRACT

Optical flow estimation is a fundamental problem in computer vision that aims to predict dense pixel-wise motion between consecutive frames [1]. However, optical flow is inherently ill-posed, since image observations only directly constrain the normal component of motion, while the full two-dimensional motion field requires additional assumptions or learned structure for recovery [1]. While real-world motion is temporal and structured, classical optical flow methods are typically formulated on pairs of frames and treat motion largely as static correspondence, limiting their ability to capture temporal coherence across short horizons [2, 3]. This project extends the classical two-frame formulation toward a short-horizon temporal modeling framework by treating motion as an evolving spatial field [2, 4]. Since both horizontal and vertical flow components can be interpreted as continuous fields whose evolution may be approximated through implicit partial differential equation (PDE)-like dynamics, we introduce a fully convolutional framework integrated with a U-shape Fourier Neural Operator (UNO) to learn these hidden motion evolution patterns from data [5, 6]. In this sense, the UNO module acts as a PDE-inspired operator learner for recognizing structured motion behavior, without requiring an explicitly specified governing equation. The framework further retains physics-informed regularizations, including edge-aware smoothness, photometric reconstruction, accumulated temporal consistency, and forward-backward consistency losses [7, 8, 9, 10]. Validated on the KITTI Scene Flow dataset [11], we evaluate three integration variants of the UNO module, demonstrating that early operator integration achieves the lowest Endpoint Error (EPE) of 2.5596. Furthermore, a downstream bounding box detection task validates that our flow estimations produce clear, stable, and physically meaningful motion regions recognizable by machine vision [12, 11].

1 Introduction

Accurate motion estimation plays a critical role in autonomous driving, robotics, and three-dimensional (3D) vision, providing information for obstacle detection, trajectory prediction, motion planning, and scene understanding [11, 13]. Optical flow estimates dense two-dimensional pixel displacement between consecutive frames, while scene flow extends this problem toward three-dimensional motion in dynamic environments [1, 4]. However, motion estimation from images is fundamentally underdetermined: image measurements directly constrain only the normal component of motion, while the remaining component requires additional assumptions, regularization, or learned priors [1].

Traditionally, optical flow is treated as a static matching problem between a pair of images [14, 15]. Classical variational methods use brightness constancy and spatial smoothness constraints, producing interpretable objectives grounded in physical assumptions [16]. However, these assumptions can fail under large displacement, occlusion, illumination change, ego-motion, independently moving objects, and discontinuous motion boundaries. Deep learning methods such as PWC-Net and RAFT improve performance through cost volumes, warping, recurrent refinement, and learned correspondence modeling [14, 15]. Recent multi-frame and spatiotemporal attention models further exploit temporal information across video sequences [3, 2]. Nevertheless, many data-driven methods still mainly treat optical flow as

a correspondence learning task, with limited explicit modeling of the physical or dynamical structure of real-world motion.

To bridge static correspondence matching and dynamic motion evolution, we propose an operator-based deep learning framework [5, 17]. Instead of predicting optical flow as isolated pairwise displacement, we model short-horizon motion as an evolving spatial field. The horizontal and vertical components of optical flow can be viewed as continuous functions over space and time, whose evolution may be approximated by implicit PDE-like dynamics. Since the true governing equation of real-world image motion is unavailable, this motivates the use of neural operators, which learn mappings between functions rather than fixed discrete data points [5].

Inspired by scientific machine learning, we introduce a U-shape Fourier Neural Operator (UNO) into a convolutional optical flow model mainline [5, 6]. The UNO module does not solve an explicitly defined PDE. Instead, it acts as a PDE-based operator learner that recognizes global spatial dependencies and temporal motion patterns from latent image-motion representations. Our proposed framework combines a PWC-style pairwise flow encoder, a joint visual-motion fusion encoder, the UNO refinement module, and a coarse-to-fine decoder [14, 15]. To evaluate the effect of operator placement, we test three integration strategies: standalone post-fusion integration, early integration after pairwise flow encoding, and late integration after visual-motion fusion.

We further incorporate weak-sense physics-informed regularization to stabilize learning and improve motion plausibility. Edge-aware smoothness preserves motion discontinuities near object boundaries, photometric and accumulated photometric losses encourage short-horizon reconstruction consistency, and forward-backward consistency penalizes geometrically inconsistent motion estimates [7, 8, 9, 10]. These constraints do not fully resolve the ill-posed nature of optical flow, yet they provide lightweight physical priors that guide the model toward more stable and interpretable motion fields. Finally, we evaluate the three UNO variants on the KITTI Scene Flow dataset and validate the physical plausibility of the predicted flows through a downstream bounding-box localization task [11, 12].

2 Related Work

2.1 Classical and Data-Driven Optical Flow

Early approaches formulate optical and scene flow estimation as variational optimization problems, where data fidelity terms are combined with regularization terms such as spatial smoothness, geometric consistency and piecewise motion assumptions [16, 1, 4]. These methods are closed connected to physical interpretations of image motion and are therefore more interpretable than purely data-driven models. However, since only the normal component of optical flow is observable, such approaches require manually designed constraints to recover the unobservable components of motion [1]. In complex driving scenes, these assumptions may become insufficient due to large displacement, occlusion, illumination changes, ego-motion, and independently moving objects [11].

Deep learning methods shifted optical flow estimation from handcrafted optimization toward learned correspondence modeling. PWC-Net uses pyramidal feature extraction, warping, and cost volumes to efficiently estimate dense flow, while RAFT performs recurrent refinement based on all-pairs field transforms [14, 15]. Recent multi-frame and attention-based methods further exploit temporal context and global spatial dependencies in video sequences [3, 2]. Although these models achieve strong performance, they often treat optical flow mainly as a discrete correspondence learning task. As a result, they may provide limited explicit modeling of the underlying physical structure of motion, especially when supervision is sparse across time [8, 9, 10].

2.2 Operator-Based Scientific Machine Learning

Scientific machine learning has introduced neural operator frameworks, such as the Fourier Neural Operator (FNO), capable of learning mappings between function spaces [5]. The family of FNOs have thus provided useful perspective for modeling global motion fields as continuous functions, rather than isolated pixels [5]. Furthermore, neural operators are designed to learn mappings between function spaces, and can approximate solution operators of partial differential equations [5]. This makes them relevant to optical flow because the horizontal and vertical flow components can be interpreted as spatial fields whose temporal evolution follows an unknown PDE-like process. However, vanilla spectral operators may lose high-frequency details that are important for optical flow, such as object boundaries, motion discontinuities, and occlusion regions. U-shaped operator architectures address this limitation by combining spectral convolutions with multi-resolution feature processing. U-FNO and U-NO preserve more local structure while maintaining global spatial modeling capacity [18, 6]. These properties motivate us to use a UNO module as a PDE-inspired operator learner for latent motion dynamics. Instead of explicitly solving a known governing equation, the module learns structured motion evolution patterns from data and complements physics-informed regularization in the loss function [5, 17, 6].

3 The Proposed Method

3.1 Problem Statement and Formulation

Instead of treating optical flow as independent pairwise correspondence, we reformulate short-horizon optical flow estimation as the learning of an evolving spatiotemporal field [1, 2]. For a two-dimensional image sequence, the optical flow can be represented as

$$u(x,t) = (u_x(x,t), u_y(x,t),$$

where x denotes spatial position, and t denotes time [1]. From the perspective of image motion, optical flow is ill-posed because only the normal component of motion is directly constrained by image observations, while the full two-component flow field requires additional structure for recovery [1]. We interpret this missing structure as an implicit motion evolution process. In other words, although the true governing equation is unknown, the flow field can be viewed as evolving according to a PDE-like operator:

$$\partial_t u = \mathcal{F}(u, I, x, t).$$

Rather than manually specifying $\mathcal{F}$, our framework learns an approximation of this operator in latent space. Given an input sequence of RGB frames

$$\{I_{t-1}, I_t, I_{t+1}, I_{t+2}\},$$

the model first extracts visual and pairwise motion features, then constructs a latent spatiotemporal representation Z_t. We approximate motion evolution using a learned temporal operator:

$$\mathcal{T}_\theta : Z_t \to Z_{t+1},$$

where $\mathcal{T}_\theta$ is implemented through the UNO module. This allows the model to learn PDE-like motion patterns from data while avoiding the need to explicitly define the physical governing equation [5, 6].

3.2 Pipeline Architecture

Our framework mainline consists of four primary modules.

1. **Pairwise Flow Encoder:** The pairwise encoder follows a PWC-style design. A shared CNN backbone extracts features from consecutive frames, and a correlation cost-volume layer computes dense correspondence between adjacent image pairs, such as I_t and I_{t+1} [14, 15]. This module provides the local pairwise motion evidence needed for optical flow estimation. To improve edge awareness, an optional Sobel-based channel can be concatenated with the RGB input, providing explicit gradient information near structural boundaries [19].
2. **Joint Visual-Motion Encoder:** The joint encoder combines raw visual information and pairwise motion embeddings. It uses a two-stream CNN structure, where one branch processes RGB images and the other processes motion embeddings [20]. These branches are fused through a time-aware, frame-stacked, two-dimensional convolution to provide short-horizon temporal modeling [3, 2].
3. **Temporal-Physics Operator Module (UNO):** The UNO module is introduced to model the evolution of latent motion representations. Since optical flow can be interpreted as a continuous field with implicit PDE-like dynamics, the UNO module acts as an operator learner that captures non-local spatial dependencies and structured temporal patterns [5, 6]. We evaluate three integration variants. The standalone variant applies UNO after visual-motion fusion and before decoding. The early integration variant applies UNO after pairwise flow encoding, allowing the operator to refine lower-level motion features. The late integration variant applies UNO after visual-motion fusion, allowing it to refine already structured spatiotemporal representations.
4. **PWC-Style Decoder:** The decoder follows a coarse-to-fine up-sampling structure to reconstruct dense optical flow from the evolved latent representation Z_{t+1} [14]. Because convolutional down-sampling can remove high-frequency boundary details, skip connections are optionally used to preserve earlier feature information and reduce excessive smoothing, following the broader idea of residual and U-Net-style feature preservation [12, 21].

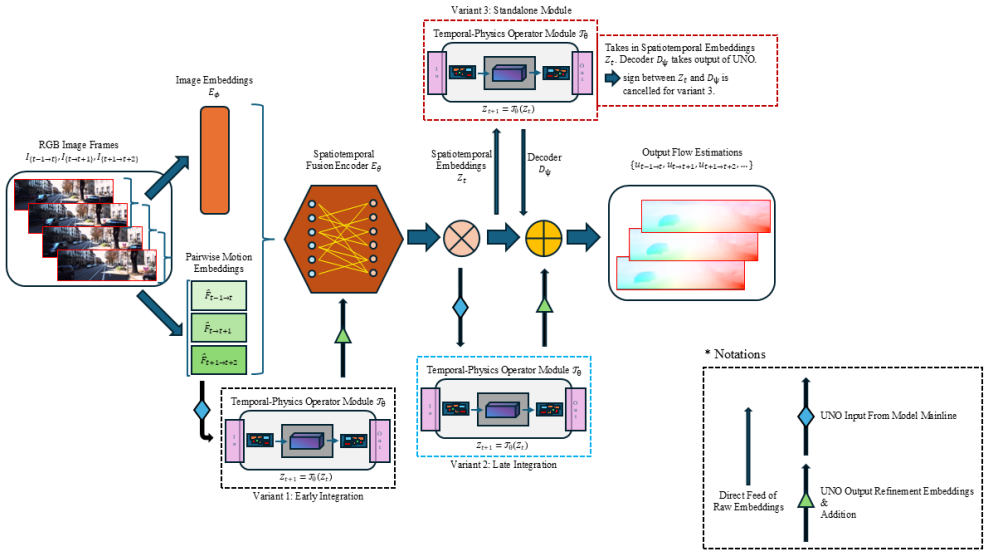


Figure 1: The 3-in-1 full short-horizon optical flow pipeline with three UNO integration variants. Dashed boxes indicate where the UNO module is inserted. The early and late variants use UNO as a residual refinement module, while the standalone variant applies UNO to the fused spatiotemporal representation before decoding.

For clarity, the overall framework is summarized in Figure 1. More detailed architecture diagrams and explanations for the three UNO integration variants are provided in Appendix A.

3.3 Loss Functions

To regularize training and enforce physical validity, we combine supervised and self-supervised constraints. The total loss is defined as:

$$\mathcal{L} = \mathcal{L}_{flow} + \lambda_{photo}\mathcal{L}_{photo} + \lambda_{acc_photo}\mathcal{L}_{acc_photo} + \lambda_{fb}\mathcal{L}_{fb} + \lambda_{smooth}\mathcal{L}_{smooth}. \tag{1}$$

Each term is formulated as follows [7, 8, 9, 10]:

- **Supervised Flow Loss ($\mathcal{L}_{flow}$):** An endpoint error (EPE) weighted by image gradient magnitude to force the model to focus on object boundaries [14, 19].

$$\mathcal{L}_{flow} = \frac{\sum_{\mathbf{x}} \|F(\mathbf{x}) - F^{gt}(\mathbf{x})\|_2 \cdot (1 + \lambda_{edge} \cdot E(\mathbf{x}))}{\sum_{\mathbf{x}} (1 + \lambda_{edge} \cdot E(\mathbf{x}))} \tag{2}$$

- **Multi-frame Photometric Loss ($\mathcal{L}_{photo}$):** A combination of L1, SSIM, and Census transforms reconstructs the target image and provides robustness against illumination variations [7, 8, 10].

$$\mathcal{L}_{photo} = \alpha\mathcal{L}_{L1} + \beta\mathcal{L}_{SSIM} + \gamma\mathcal{L}_{census} \tag{3}$$

- **Accumulated Photometric Loss ($\mathcal{L}_{acc_photo}$):** This term extends photometric supervision across multi-step frame transitions by composing consecutive flows and accumulating reconstruction penalties over longer temporal horizons [9, 3].

$$F_{t\to t+2}(\mathbf{x}) = F_{t\to t+1}(\mathbf{x}) + F_{t+1\to t+2}\left(\mathbf{x} + F_{t\to t+1}(\mathbf{x})\right) \tag{4}$$

$$\mathcal{L}_{acc_photo} = \frac{\sum_{\mathbf{x}} \mathcal{L}_{photo}^{(t,t+2)}(\mathbf{x}) \cdot W_{t,t+2}(\mathbf{x})}{\sum_{\mathbf{x}} W_{t,t+2}(\mathbf{x})} \tag{5}$$

- **Forward-Backward Consistency ($\mathcal{L}_{fb}$):** This term penalizes geometric inconsistencies when mapping pixels forward and backward in time [8, 9].

$$\mathcal{L}_{fb} = \sum_{\mathbf{x}} \|F_{fw}(\mathbf{x}) + F_{bw}\left(\mathbf{x} + F_{fw}(\mathbf{x})\right)\|_1 \tag{6}$$

- **Edge-Aware Smoothness Loss ($\mathcal{L}_{smooth}$):** Extended to all predicted flows in the sequence, this gradient-weighted penalty encourages smoothness in homogeneous regions while preserving discontinuities at structural boundaries [7, 19, 10].

$$\mathcal{L}_{smooth} = \frac{1}{T-1} \sum_{t=1}^{T-1} \sum_{\mathbf{x}} |\nabla F_t(\mathbf{x})| \cdot e^{-|\nabla I_t(\mathbf{x})|} \tag{7}$$

4 Performance Evaluation

4.1 Experimental Setup

Models were trained and evaluated on the KITTI Scene Flow dataset, which contains real-world driving scenes with dynamic objects, camera motion, depth variation, and complex geometric structure [11]. To isolate the effect of operator placement, we held the main hyperparameters, backbone structure, decoder design, and training configuration constant across the three UNO variants. The only major architectural difference is the stage at which the UNO module is inserted into the pipeline. We evaluate performance using standard optical flow metrics: End-Point Error (EPE), per-image EPE (F1-EPE), and the optical flow outlier ratio (F1-all%) [11, 14, 15]. Lower values indicate better performance for all three metrics.

4.2 Quantitative Results and Training Dynamics

As demonstrated in Table 1, the integration stage of the UNO module has a clear influence on model behavior. The Early Integration variant achieves the best average accuracy, with the lowest EPE and F1-EPE values. This suggests that allowing the operator to act on lower-level pairwise motion features can improve correspondence alignment before visual-motion fusion. In this setting, the UNO module is closer to the source of motion representation and can refine features before they are compressed into more structured latent embeddings [14, 5, 6].

Conversely, the Original Standalone variant achieves the lowest F1-all% at 20.59%, indicating fewer large outlier errors. This result suggests that the standalone operator produces more conservative motion updates. Since it operates after the visual-motion fusion stage, it receives more structured spatiotemporal embeddings and may therefore avoid aggressive corrections that introduce large local errors. The Late Integration variant performs between the other two settings, suggesting that post-fusion refinement is useful but also causes stronger latent-space intervention than the standalone configuration [5, 18, 6].

Model Variant	**EPE ↓**	**F1-EPE ↓**	**F1-all% ↓**
Standalone (Original)	2.9234	2.7889	**20.59**
Early Integration	**2.5596**	**2.5596**	22.11
Late Integration	2.7647	2.6673	23.30

Table 1: Quantitative evaluation of UNO integration variants on the KITTI dataset.

The training dynamics in Figure 2 further explain these differences. All three variants show decreasing weighted EPE loss, indicating that the baseline supervised flow objective is learnable under each integration strategy. However, the Late Integration variant produces the largest latent space changes. This suggests that when UNO is applied after visual-motion fusion, the operator acts more strongly as a refinement module and directly modifies already structured fused representations. In contrast, the Early Integration and Standalone variants show smaller latent changes, indicating more conservative updates.

The temporal consistency trend provides a complementary observation. The Standalone variant achieves the lowest temporal consistency loss, while the Early Integration variant produces the highest temporal consistency loss. This suggests that applying UNO too early can improve average pixel-wise correspondence but may also introduce short-horizon temporal fluctuation because lower-level features are less structured. Therefore, early integration is more effective for average EPE reduction, while standalone integration better preserves temporal smoothness. This tradeoff supports the central interpretation of our project: neural operators can model global motion structure, but their effectiveness depends strongly on the resolution and semantic maturity of the features they receive [5, 6].

Visually, the predicted flows in Figure 3 show that all three variants capture the dominant moving regions, especially vehicles and other foreground objects. The use of edge-aware smoothness helps preserve object boundaries by reducing excessive smoothness penalties near strong image gradients [7, 8]. At the same time, the results also reveal a limitation

of the current architecture: convolutional down-sampling compresses the feature maps before operator processing, which can remove high-frequency details important for sharp motion boundaries. This explains why the UNO module improves structural consistency but does not fully eliminate over-smoothing introduced by CNN-based up-sampling [21, 18]. The short-horizon samples in Figure 4 further show that the predicted flows remain broadly coherent across adjacent frame transitions, suggesting that the model learns temporal motion structure beyond independent pairwise matching. However, remaining fluctuations near object boundaries and low-texture regions indicate that stronger multi-frame supervision or higher-resolution operator inputs may be needed for more stable temporal evolution.

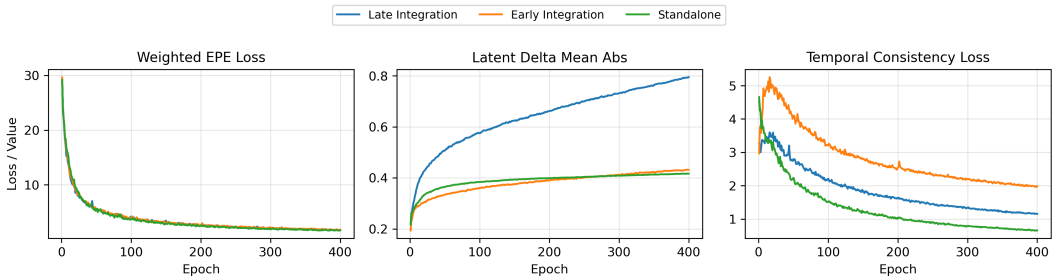


Figure 2: Training dynamics comparing the three UNO integration strategies across weighted EPE, latent space changes, and temporal consistency.

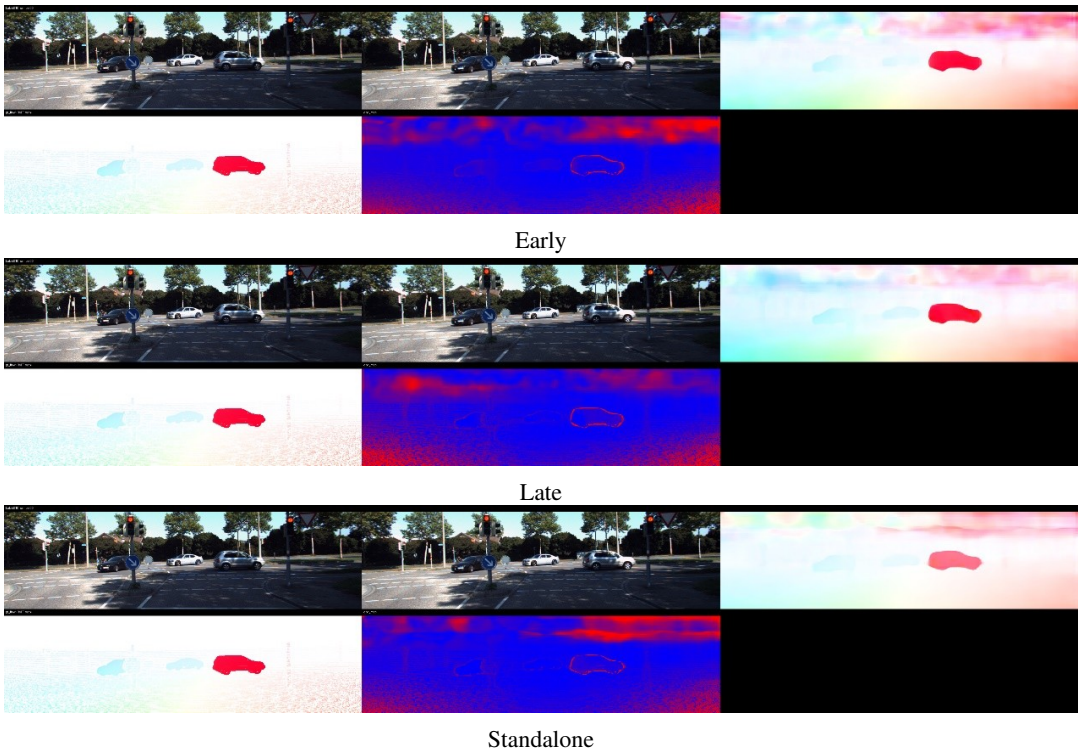


Figure 3: Pairwise optical flow estimation across three UNO integration variants.

4.3 Downstream Task: Motion Recognition

Beyond pixel-level evaluation, we further examine whether the generated optical flows contain motion structures that can be recognized by a machine vision model. This downstream task is not intended to introduce a new object detector. Instead, it serves as a diagnostic test for the physical and structural meaningfulness of the predicted flow fields. If the estimated flows preserve coherent moving regions, then a lightweight detector should be able to localize objects based primarily on motion cues [11, 22].

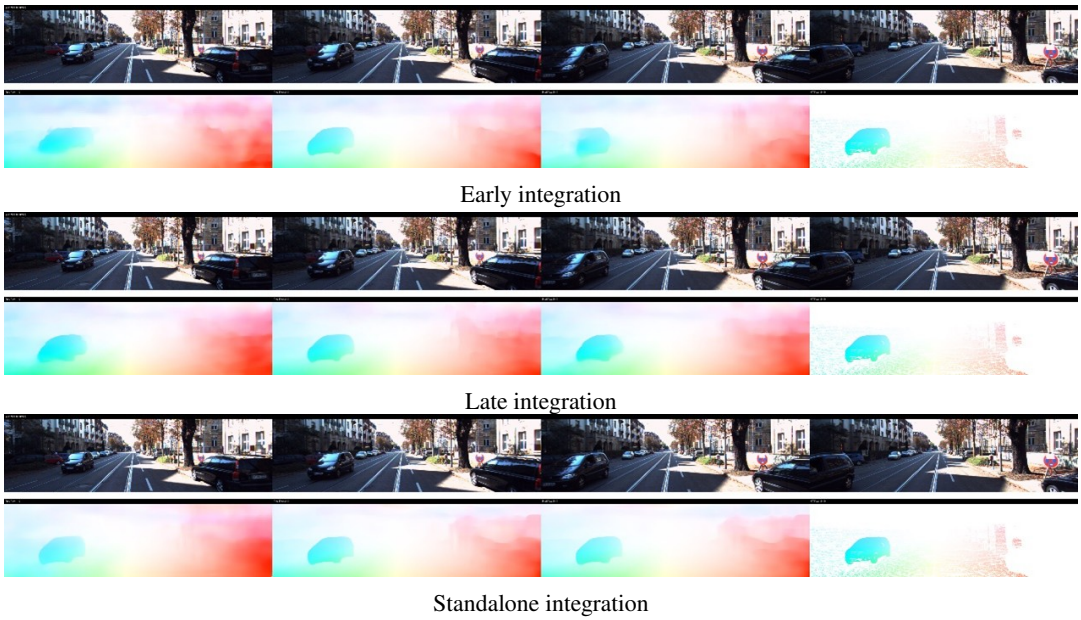


Figure 4: Short-horizon flow prediction samples across the three UNO integration variants. The results show predicted motion fields across consecutive frame transitions, illustrating temporal coherence beyond single pairwise estimation.

We implement this test using a ResNet50-based detection backbone modified to accept flow-based inputs. Since optical flow contains horizontal and vertical displacement channels, we concatenate an additional one-channel bounding-box mask during training, resulting in a three-channel input representation. We compare two training settings: one using ground-truth flow as detector input and another using estimated flow as detector input. The ground-truth-flow setting provides cleaner motion labels but creates a train-test mismatch when inference is performed on estimated flows. The estimated-flow setting reduces this distribution shift because the detector is trained on inputs closer to those used during inference [12, 14, 15].

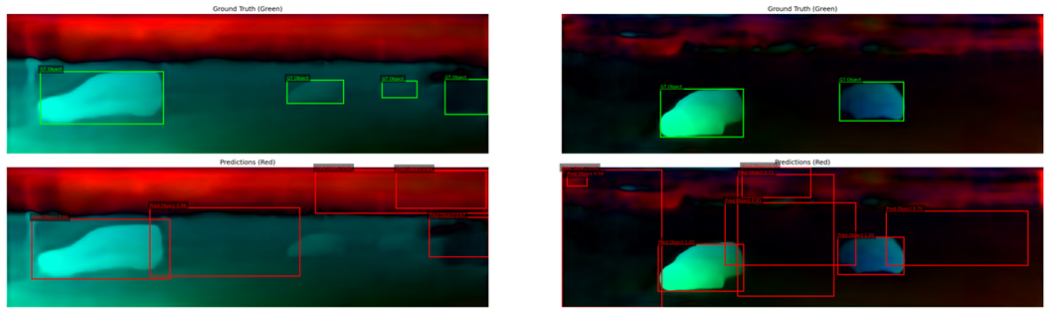
The downstream results show that training the detection model on estimated optical flows improves visual inference by reducing out-of-distribution effects. As shown in Figure 5, the detector is able to identify major moving regions with relatively few false positives, suggesting that the predicted flow fields contain recognizable and stable motion structures. This supports the claim that our optical flow model does not merely minimize pixel-wise error, but also produces motion representations useful for higher-level perception. Such behavior is important for autonomous driving and robotic vision, where optical flow is often used as an intermediate representation for object localization, motion understanding, and scene interpretation [11, 13, 22].

At the same time, this downstream task also reveals a limitation. Detection performance depends strongly on the quality and distribution of the flow input. When the detector is trained on ground-truth flow but evaluated on estimated flow, background ego-motion and prediction artifacts can create false positives. This indicates that future work should consider joint training or domain adaptation between flow estimation and downstream perception. Overall, the result confirms that flow quality directly affects high-level machine perception performance.

5 Conclusion and Future Work

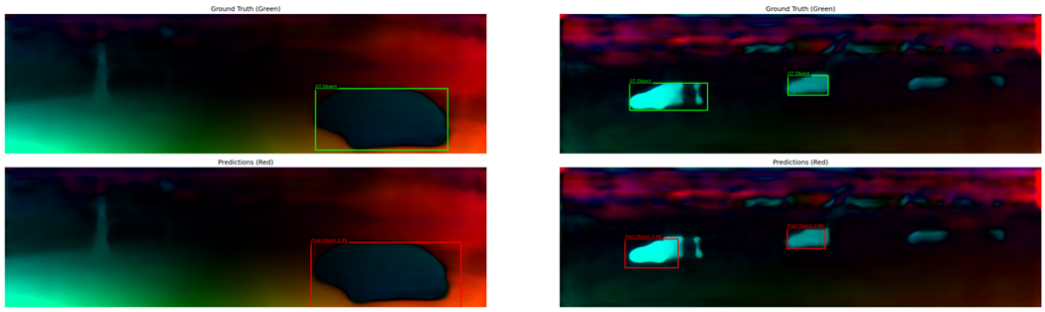
In this work, we reformulated short-horizon optical flow estimation from a static correspondence task into an evolving spatiotemporal motion modeling problem. Starting from the observation that optical flow is inherently ill-posed and only partially observable from image measurements, we introduced an operator-based framework to learn hidden motion evolution patterns from short RGB sequences [1, 4]. By integrating a U-shape Fourier Neural Operator (UNO) with physics-inspired loss functions, the model captures both structural motion boundaries and temporal motion patterns without requiring an explicitly defined governing PDE [5, 6, 7, 8].

Among the three architectural variants, early UNO integration achieves the best pixel-wise accuracy, suggesting that applying the operator closer to pairwise motion features improves average correspondence alignment. In contrast, the standalone UNO variant produces the lowest outlier ratio and strongest temporal consistency, indicating that operating on more structured fused representations leads to more conservative and temporally stable predictions. Furthermore,



(a) (b)

Trained on ground-truth flow



(a) (b)

Trained on estimated flow

Figure 5: Downstream motion-recognition results. Green boxes show ground-truth labels and red boxes show predictions. Training on estimated flows reduces the train-test mismatch and produces cleaner localization.

regardless of UNO's claimed capability of preserving higher frequency details, the results on our standalone variant still show that it favors more global information compared to the convolutional mainline [6]. These tradeoff show that neural operators can contribute to optical flow estimation, but their effectiveness depends strongly on their means of integration, and how much spatial detail is provided for operator processing.

The downstream localization task further supports the physical meaningfulness of the predicted flow fields. Although the detector is not intended as a standalone object detection contribution, its ability to identify major motion regions from estimated flows suggests that the generated motion fields contain recognizable structures useful for higher-level machine perception [11, 22]. At the same time, the results show that downstream performance is sensitive to train-test mismatch between ground-truth and estimated flow inputs, confirming that flow quality directly affects later perception tasks.

Future work may include extending the current pipeline from two-dimensional optical flow to three-dimensional scene flow prediction using KITTI disparity data. This direction is especially relevant because neural operators can model motion as a continuous spatiotemporal field and may help approximate unobserved motion components through learned structural regularities [11, 1, 4]. Another important direction is to preserve higher-resolution information before operator processing through stronger skip connections, multi-scale operator inputs, or reduced down-sampling; or instead, to enable FNO-based models to learn, recognize, and preserve useful higher frequency signals without filtering them out. Finally, as progress has been continuously emerging in the field of optical estimation, integrating stronger, state-of-the-art correspondence models into the temporal-physics operator framework may also further improve accuracy in weakly supervised or unsupervised settings [15, 3, 10].

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A Architecture Specifications

This appendix provides enlarged architecture diagrams for the three UNO integration variants used in the proposed short-horizon optical flow framework. The variants differ mainly in where the Temporal-Physics Operator Module is inserted relative to pairwise motion encoding, spatiotemporal fusion, and final flow decoding.

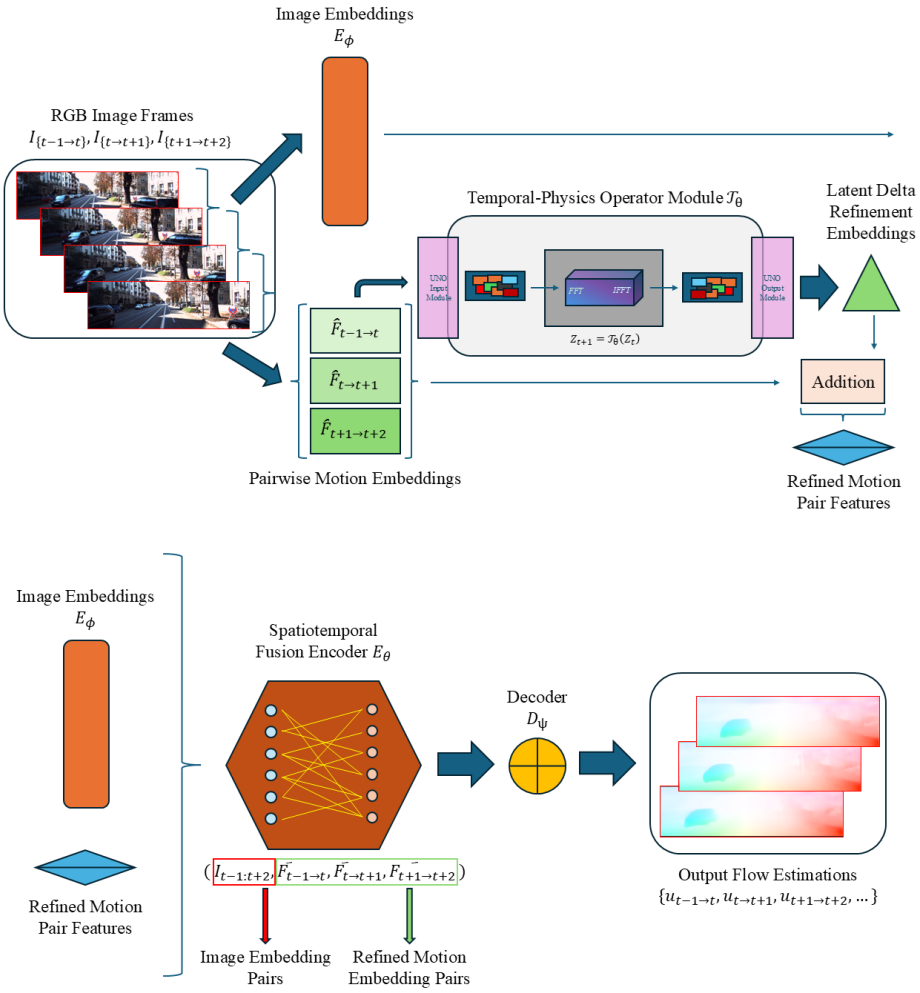

Figure 6: Early integration architecture. The UNO-based Temporal-Physics Operator Module is inserted after pairwise motion embedding extraction and before final visual-motion fusion. In this setting, the operator acts on relatively low-level pairwise motion features and produces latent delta refinement embeddings, which are added back to the motion representation before decoding. This design allows operator-based refinement to influence motion features before they are compressed into later spatiotemporal embeddings.

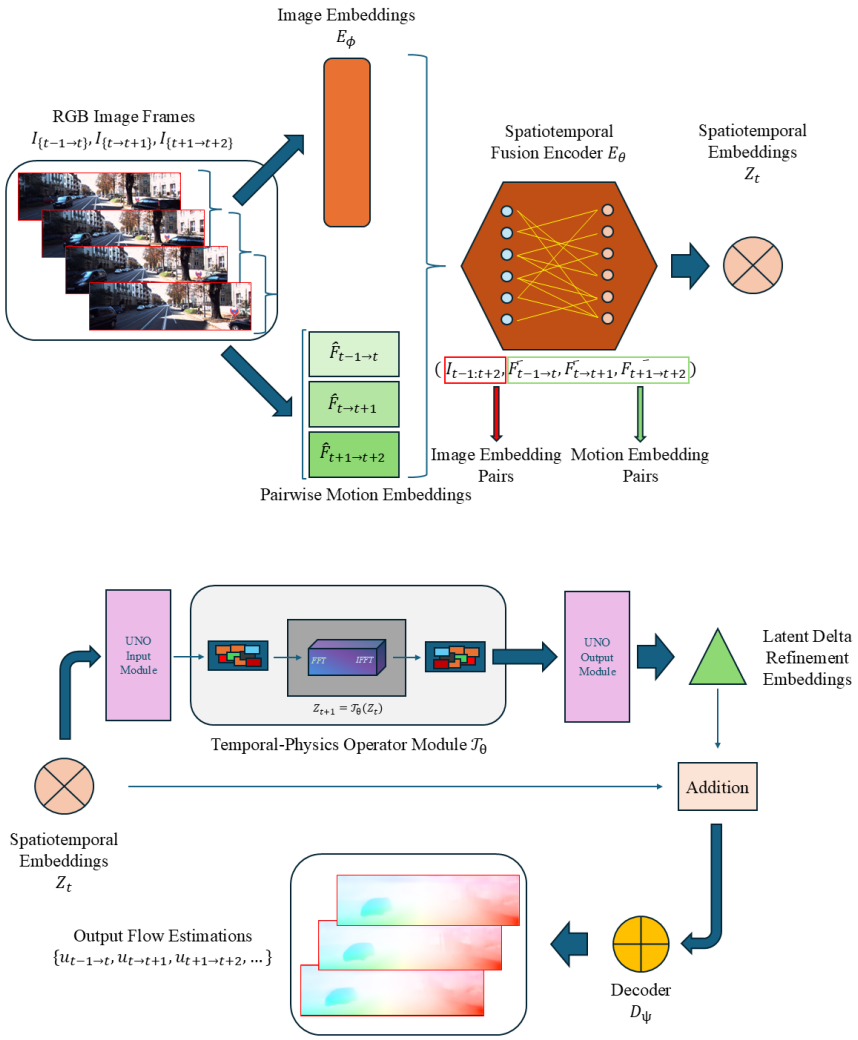

Figure 7: Late integration architecture. RGB image embeddings and pairwise motion embeddings are first combined by the spatiotemporal fusion encoder to form the latent representation Z_t. The UNO-based Temporal-Physics Operator Module is then applied after fusion as a residual refinement stage. The resulting latent delta refinement embeddings are added to the fused representation before being passed to the decoder. This variant allows the operator to refine already structured spatiotemporal features.

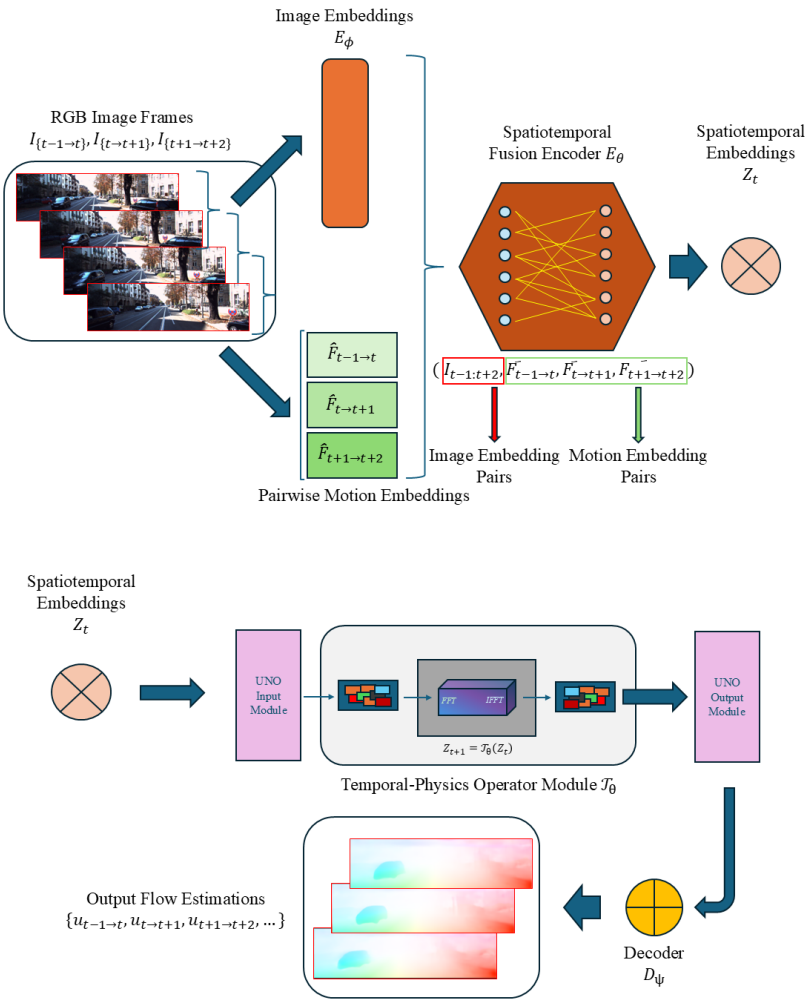

Figure 8: Standalone UNO architecture. The model first constructs spatiotemporal embeddings Z_t through image-motion fusion, and then feeds this representation into the UNO-based Temporal-Physics Operator Module as an independent operator block. The refined output is decoded into short-horizon optical flow estimations. Compared with the residual early and late variants, this configuration treats the operator module as a standalone post-fusion motion evolution learner.